A plasmid called the fertility or F factor plays a major role in conjugation. The F factor is about 100 kilobases long and bears genes responsible for cell attachment and plasmid transfer between specific bacterial strains during conjugation. F factor is made up of

a. tra region (tra operon / transfer genes): genes coding the F pilus and DNA transfer,

b. Insertion sequence: genes assisting plasmid integration into host cell chromosome.

Thus, the F factor is an **episome** - a genetic material that can exist outside the bacterial chromosome or be integrated into it.

During $F^+ \times F^-$ mating or conjugation (Figure 12.17a) the F factor replicates by the rolling circle mechanism and a copy moves to the recipient. The channel for DNA transfer could be either the hollow F pilus or a special conjugation bridge formed upon contact. The entering strand is copied to produce double – stranded DNA.

F factor can integrate into the bacterial chromosome at several different locations by recombination between homologous insertion sequences present on both the plasmid and host chromosomes. The integration of F factor into bacterial chromosome results in formation of HFR (High Frequency Recombination) cell. When integrated, the Fplasmid's tra operon is still functional; the plasmid can direct the synthesis of pili, carry out rolling circle replication, and transfer genetic material to an F^- recipient cell. An HFR cell is so called because it exhibits a very high efficiency of chromosomal gene transfer in comparison with F^+ cells. In F^+ cells the independent F factor rarely transfer chromosomal genes hence the recombination frequency is low. Figure 12.17b shows formation of HFR cell. When an HFR cell is mated with F^- cell the F^- recipient does not become F^+ unless the whole chromosome is transferred as explained in Figure 12.17c. The connection usually breaks before this process is finished. Thus, complete F factor usually is not transferred, and the recipient remains F^-.

Because the F plasmid is an episome, it can leave (deintegrate) the bacterial chromosome. Sometimes during this process, the plasmid makes an error in excision and picks up a portion of the chromosomal material to form an F′ plasmid. Figure 12.17d shows formation of F′. During $F'XF^-$ conjugation (Figure 12.17e) the recipient becomes F′ and is a partially diploid since it has two set of the genes carried by the plasmid.

The natural phenomenon of conjugation is now exploited for gene transfer and Recombinant DNA technology. In general, the plasmids lack conjugative functions and therefore, they are not as such capable of transferring DNA to the recipient cells. However, some plasmids with conjugative properties can be prepared and used.

12.7.3 Transduction

Transduction is the transfer of bacterial genes from one bacteria to other by viruses. Example: Bacteriophage (Bacterial viruses). To understand the

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